



Problem Statement 5: Enterprise Project Management & Team Collaboration Platform (Jira + Notion + Slack + GitHub)

Difficulty

Advanced

Domain

SaaS • Productivity • Project Management • DevTools

Duration

24–48 Hours

Background

Modern software teams use multiple tools like Jira for issue tracking, Notion for documentation, Slack for communication, GitHub for development, Google Drive for files, and spreadsheets for reporting. This fragmentation causes context switching, duplicate work, poor visibility, and inefficient collaboration.

Build a production-ready Project Management & Team Collaboration Platform that combines project planning, task management, documentation, sprint planning, real-time collaboration, issue tracking, analytics, and AI-powered productivity into a single application.

Project management platforms remain one of the most common enterprise SaaS products because they require sophisticated role management, workflows, permissions, notifications, and integrations while solving a genuine business need.

Objective

Develop a full-stack SaaS application that enables organizations to manage

- Projects
- Teams
- Tasks
- Sprints

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- Documentation
 - Meetings
 - Files
 - Issues
 - Time Tracking
 - Analytics
 - Team Communication
-

Mandatory Tech Stack

Frontend

- Next.js / React
- TypeScript
- Tailwind CSS
- Responsive Design

Backend

- Node.js
- Express / Next.js API

Database

- PostgreSQL / MongoDB

Authentication

- Google OAuth
- Email Authentication
- JWT / Secure Sessions

Storage

- AWS S3 / Cloudinary

Deployment

- Vercel
 - Railway
 - Docker (Bonus)
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Authentication Requirements

Support

Sign Up

- Google OAuth
 - Email Registration
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Login

- Google Login
 - Email Login
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Email Verification

Mandatory before workspace creation.

Forgot Password

Support

- OTP
OR
 - Password Reset Email
-

Session Management

- Secure Cookies
 - Refresh Tokens
 - Multiple Device Login
 - Logout from All Devices
-

Two-Factor Authentication (Bonus)

Available for

- Workspace Owners

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- Admins
-

User Roles

Workspace Owner

Can

- Create Workspace
 - Invite Members
 - Manage Billing
 - Create Projects
 - Assign Roles
 - View Analytics
 - Manage Integrations
-

Project Manager

Can

- Create Projects
 - Create Sprints
 - Assign Tasks
 - Approve Work
 - Generate Reports
 - Manage Milestones
-

Team Member

Can

- View Assigned Tasks
- Update Status
- Upload Files
- Comment
- Log Time
- Create Subtasks



Client

Can

- View Progress
- View Shared Documents
- Approve Deliverables
- Comment on Tasks

Cannot

- Modify Projects
-

Admin

Platform-wide access

Manage

- Users
 - Organizations
 - Billing
 - Permissions
 - Audit Logs
 - System Settings
-

Landing Page

Include

- Hero Section
- Product Features
- Pricing
- Testimonials
- FAQ
- Contact
- Blog
- Footer
- Responsive Navigation
- Dark Mode



- SEO
-

Workspace Module

Users should be able to

- Create Workspace
 - Upload Logo
 - Invite Members
 - Join via Invite Link
 - Manage Departments
 - Configure Permissions
-

Dashboard

Display

- Active Projects
 - Pending Tasks
 - Upcoming Deadlines
 - Sprint Progress
 - Team Activity
 - Notifications
 - Calendar
 - Productivity Score
-

Project Module

Each project contains

- Name
- Description
- Status
- Priority
- Deadline
- Team Members
- Milestones
- Files



- Wiki
 - Activity Timeline
-

Task Management

Create tasks with

- Title
- Description
- Assignee
- Reporter
- Labels
- Priority
- Status
- Due Date
- Attachments
- Checklists
- Story Points

Support

- Subtasks
 - Dependencies
 - Recurring Tasks
 - Bulk Actions
-

Kanban Board

Support drag-and-drop workflow

Backlog



To Do



In Progress





Code Review



Testing



Completed

Allow users to customize columns.

Sprint Management

Create

- Sprint Name
- Goal
- Duration
- Capacity
- Sprint Backlog

Display

- Burndown Chart
 - Velocity
 - Sprint Completion
 - Blockers
-

Documentation (Wiki)

Each project should include a collaborative documentation system with

- Rich Text Editor
- Markdown Support
- Code Blocks
- Tables
- Images
- Version History
- Search



- Nested Pages
-

Calendar

Display

- Deadlines
 - Meetings
 - Sprint Dates
 - Holidays
 - Team Availability
-

File Management

Upload

- Images
- PDFs
- Videos
- ZIP Files
- Documents

Support

- Folder Structure
 - Preview
 - Download
 - Version History
-

Team Chat

Provide

- Project Channels
- Direct Messages
- Threaded Conversations
- Mentions (@user)
- Emoji Reactions
- File Sharing

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Bonus

- WebSocket-powered real-time messaging
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Meeting Module

Create meetings with

- Agenda
- Participants
- Notes
- Action Items
- Calendar Invites

Store meeting history.

Time Tracking

Members can

- Start Timer
- Stop Timer
- Manual Log
- Daily Work Log

Managers can view

- Productivity Reports
 - Hours by Project
 - Team Utilization
-

Notifications

Real-time notifications for

- Task Assigned
- Task Completed
- Deadline Reminder
- Mention

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- File Uploaded
 - Comment Added
 - Sprint Started
 - Sprint Ended
-

Search

Global Search

Search

- Projects
 - Tasks
 - Members
 - Documents
 - Messages
 - Files
-

Analytics Dashboard

Display

- Project Progress
 - Sprint Velocity
 - Task Completion Rate
 - Team Productivity
 - Time Logged
 - Workload Distribution
 - Burnup/Burndown Charts
 - Project Health Score
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Integrations (Bonus)

Support integration with

- GitHub
- Google Calendar
- Google Drive
- Slack

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- Discord
 - Zoom
 - Microsoft Teams
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AI Features (Bonus)

Implement one or more of the following:

- AI task prioritization
 - AI sprint planning suggestions
 - AI meeting summarization
 - AI documentation generator
 - AI project risk prediction
 - AI workload balancing
 - AI task estimation
 - AI chatbot for workspace queries
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Admin Panel

Manage

- Organizations
 - Users
 - Roles
 - Workspaces
 - Projects
 - Billing Plans
 - Reports
 - Integrations
 - Audit Logs
 - Feature Flags
-

UI/UX Requirements

The application must include

- Responsive Design
- Mobile-First Layout



- Dark & Light Theme
 - Drag-and-Drop Boards
 - Skeleton Loading
 - Empty States
 - Toast Notifications
 - Keyboard Shortcuts
 - Command Palette (Ctrl/Cmd + K)
 - Smooth Animations
 - Accessibility (WCAG basics)
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Security Requirements

Implement

- Password Hashing (bcrypt/Argon2)
 - Role-Based Access Control (RBAC)
 - Organization-level Permission System
 - Input Validation
 - XSS Protection
 - CSRF Protection (where applicable)
 - Rate Limiting
 - Secure File Upload Validation
 - Audit Logging
 - API Authorization Middleware
 - Secure Environment Variable Management
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Database Entities

Minimum tables/collections

- Users
- Organizations
- Workspaces
- Roles
- Permissions
- Projects
- Sprints
- Tasks
- Subtasks
- Labels



- Comments
 - Attachments
 - Wiki Pages
 - Meetings
 - Time Logs
 - Notifications
 - Messages
 - Activity Logs
 - Integrations
 - Settings
-

Bonus Features

- Gantt Chart
 - Mind Map View
 - Whiteboard Collaboration
 - Offline Mode (PWA)
 - Multi-language Support
 - Public Project Pages
 - Guest Access
 - Team Performance Leaderboard
 - Calendar Sync
 - Email Digests
 - Dockerized Deployment
 - CI/CD Pipeline
 - API Documentation (Swagger/OpenAPI)
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Judging Criteria

Criteria	Weight
Core Functionality	30%
UI/UX & User Experience	20%
Code Quality & Architecture	15%
Authentication & Security	10%
Database Design	10%

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Performance & Scalability 5%

Innovation & AI Features 5%

Deployment & Documentation 5%

Expected Deliverables

- Public GitHub repository with clean commit history
- Live deployed application
- README with setup and architecture details
- API documentation (Swagger/OpenAPI)
- ER Diagram / Database Schema
- Demo video (3-5 minutes)
- Test accounts for every role
- `.env.example` file
- Docker setup (Bonus)
- CI/CD workflow (Bonus)